WHAT PROBLEMS DO TALTECH'S BEST RESEARCH ARTICLES SOLVE?

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Edited by: Kaire Uusen

TalTech's best scientific articles of the year were revealed, including one in the social and humanities, two in technology, and one in the natural, exact, and health sciences. The award-winning articles range in topics from medicine and new technologies to rethinking AI and human collaboration.

BEST ARTICLE IN THE SOCIAL AND HUMANITIES FIELD: RETHINKING HUMAN-AI COLLABORATION

Why doesn't human-AI collaboration always work as expected? This question is raised and answered in Mari-Klara Stein's research article , which was selected as the best of the year in the social sciences and humanities and was published in the journal Information and Organization.

Thus, Stein's scientific article offers an original approach to one of today's most pressing questions about human-AI cooperation, which, unfortunately, does not always achieve the promised trust and performance. According to the jury, Stein's article contains an original conceptual rethinking of the limits of human-AI cooperation. The central innovation of the article also lies in the critical rethinking of Herbert Simon's concept of bounded rationality, based on contemporary developments in both artificial intelligence `nologies and the study of human-AI cooperation. "The article shows that the classical assumptions of bounded rationality - in

particular the understanding of cognition as mere information processing - are no longer sufficient to explain the interaction between humans and AI in organizational and societal contexts," the jury emphasizes. Thus, Stein's research is an important contribution to the development of human-centered and trustworthy AI, which has a broader impact on both science and policymaking. The article offers an original and theoretically ambitious contribution to the discussion about rationality and intelligence in the age of artificial intelligence.

Stein, Mari-Klara; Shollo, Arisa (2025). Microfoundations of rationality in the age of AI_On emotions, bodies and intelligence. Information and Organization, 35, #100583

TWO OF THE BEST ARTICLES IN THE FIELD OF TECHNOLOGY: THE ULTRA-FAST FITSA METHOD AND ARTIFICIAL INTELLIGENCE IN THE BATTERY CHARGING PROCESS

In the field of technology, two of the best were selected: a scientific article published in the journal Science Advances on the development of a new analysis method FITSA and a scientific article published in the journal IEEE Transactions on Intelligent Vehicles on a methodology based on neural networks for controlling and evaluating battery charging processes.

Ultrafast FITSA method

According to the jury, the article introducing the FITSA method is a landmark scientific achievement that solves a long-standing and fundamental problem in the study of molecular processes and opens up new possibilities for the quantitative analysis of both wildlife and technological systems. More specifically, Professor Marko Vendelin's team developed a completely new method for studying molecules that can make drug development much faster and cheaper than before. It is the FITSA method, which means statistical analysis of fluorescence intensity time series. While the FCS method, which has been used for decades, was slow and required a huge amount of data, now thanks to FITSA, Estonian scientists have found a way to do the same work a thousand times faster and more reliably. It is also interesting to compare that 300 to 21,000 times less (!) data is needed to evaluate molecular properties using FITSA compared to classical FCS to obtain the same result. Thanks to this, FITSA opens up new possibilities for studying processes that were previously impossible.

Karimi, H.; Laasmaa, M.; Pihlak, M.; Vendelin, M. (2025). Statistical analysis of fluorescence intensity transients with Bayesian methods. Science Advances.

Artificial intelligence streamlines the battery charging process

The authors of the second best article in the field of technology are TalTech researchers Gilbert Zequera, Viktor Rjabtšikov, Anton Rassõlkin, Toomas Vaimann and Ants Kallaste, whose research systematically combines data-driven artificial intelligence, virtual sensors and a digital twin, thus enabling more accurate charge prediction, adaptive control and real-time battery aging calculation. The scientific article introduces a robust deep learning method that uses neural networks in the TensorFlow environment and Python as the programming language. The goal is to predict battery open-circuit voltage (OCV) and improve battery health assessment.

According to the jury, the solution proposed in the research paper offers a clear scientific and technological innovation in the control of energy storage systems. The integration of deep learning models into the control of energy systems and the use of artificial intelligence-based virtual sensors in the optimization of charging strategies were highlighted.

All authors are researchers at Tallinn University of Technology, and the university's contribution to the work is leading. The article is directly related to the ETAG start-up grant PSG453 "Digital twin of the drivetrain of an autonomous electric vehicle" (2020–2024) and represents one of the most important scientific outputs of the project. The results of the work are directly applicable to the management of the energy systems of electric vehicles, helping to improve their safety, reliability and energy efficiency.

Gilbert Zequera, RA; Ryabchikov, V.; Rassölkin, A.; Vaimann, T.; Kallaste, A. (2025). Deep Learning methodology for charging management applications in battery cells based on Neural Networks. IEEE Transactions on Intelligent Vehicles.

THE BEST SCIENTIFIC ARTICLE IN THE FIELD OF NATURAL, EXACT AND HEALTH SCIENCES: NOVEL POSSIBILITIES IN THE TREATMENT OF THE BRAIN AND SPINAL CORD

The best scientific article of the year was written with the participation of Indrek Koppel, a junior professor at the Institute of Chemistry and Biology. It was published in the high-profile journal Cell and examined the response of cells in the peripheral nervous system to injuries.

The peripheral nervous system includes all the nerves that connect the brain and spinal cord to the rest of the body. When researchers observed how cells in the peripheral nervous system respond to injury, they unexpectedly discovered that repetitive DNA sequences, previously thought to be useless, actually play an important role in repairing nerve damage. The discovered mechanism may open up new possibilities for treating brain and spinal cord injuries in the future. The experiments also revealed that the function that supports nerve cell regeneration can be turned off, and the researchers also investigated whether the same happens in the central nervous system. “This ability would be particularly important because the body recovers significantly worse from brain and spinal cord injuries than from other body injuries,” explained Indrek Koppel on the Novaator science portal.

The jury also emphasized the value of the scientific article, highlighting that Cell is a journal of absolute scientific excellence, with an impact factor comparable to such top journals as Nature and Science. The work was completed in collaboration with several research groups, mainly in the United States, and under the leadership of the working group of Mike Fainzilber, professor of molecular neurobiology at the Weizmann Institute of Science in Israel.

Zahavi, Eitan Erez; Koppel, Indrek; Kawaguchi, Riki; Oses-Prieto, Juan A.; Briner, Adam; Monavarfeshani, Aboozar; Dalla Costa, Irene; van Niekerk, Erna; Lee, Jinyoung; Matoo, Samaneh; Hegarty, Shane; Donahue, Ryan J.; Sahoo, Pabitra K.; Ben-Dor, Shifra; Feldmesser, Ester; Ryvkin, Julia; Leshkowitz, Dena; Perry, Rotem Ben-Tov; Cheng, Yuyan; Farber, Eli; Abraham, Ofri; Samra, Nitzan; Okladnikov, Nataliya; Alber, Stefanie; Albus, Christine A.; Rishal, East; Ulitsky, Igor; Tuszynski, Mark H.; Twiss, Jeffery L.; He, Zhigang; Burlingame, Alma L.; Fainzilber, Mike (2025) Repeat-element RNAs integrate a neuronal growth circuit. Cell 188, 4350-4365.e22 (PRG2206)

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10.03.2026

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